

## EZIPAY KENTRO API

The Kentro platform is designed to provide centralized API access to Ezipay services across Africa. The APIs are developed based on the REST architecture.

### 1. TRANSACTION REQUESTS

In order to initiate a transaction on the Kentro platform, an HTTP POST request has to be made to the service manager platform for processing and forwarding. The sequence for processing is as follows:

- a. Post transaction details to Kentro platform
  - b. Transaction details are verified and errors returned if any.
  - c. Transaction is processed where there is no error and status returned.
  - d. Client account is debited to reflect transaction changes.
- In the case of mobile money transactions, final transaction status is provided via details pushed to a provided callback url.

#### REQUEST

```
{
  "apiKey": "string", M
  "service": "string", M
  "serviceType": "string", M
  "channel": "string", M
  "invoiceNo": "string", M
  "customerRequest": {
    "senderFirstName": "string", M
    "senderMiddleName": "string", O
    "senderLastName": "string", M
    "senderGender": "string", M
    "senderAddress": "string", M
    "senderCity": "string", M
    "senderState": "string", M
    "senderZipCode": "string", M
    "senderCountry": "string", M ISO 3166-1 alpha-2
    "senderMobile": "string", M
    "senderNationality": "string", M
    "senderIdType": "string", M (P, Passport, D, Drivers License, N, National ID)
    "senderIdNumber": "string", M
    "senderIdIssueCountry": "string", M ISO 3166-1 alpha-2
    "senderIdIssueDate": "string", M
    "senderIdExpireDate": "string", M
    "senderDateOfBirth": "string", O
    "senderOccupation": "string", O
  }
}
```

```

"senderSourceOfFund": "string", M
"senderSecondaryIdType": "string", 0
"senderSecondaryIdNumber": "string", 0
"senderEmail": "string", M
"purposeOfRemittance": "string", M
"receiverFirstName": "string", M
"receiverMiddleName": "string", 0
"receiverLastName": "string", M
"receiverAddress": "string", M
"receiverContactNumber": "string", M
"receiverState": "string", M
"receiverAreaTown": "string", 0
"receiverCity": "string", M
"receiverCountry": "string", M ISO 3166-1 alpha-2
"receiverIdType": "string", M (P, Passport, D, Drivers License, N, National ID)
"receiverIdNumber": "string", M
"receiverIdExpiry": "string", M
"receiverDob": "string", 0
"receiverIdCountry": "string", M ISO 3166-1 alpha-2
"receiverZipCode": "string", M
"receiverEmail": "string", M
"receiverAccountType": "string", 0
"transferAmount": "string", M
"remitCurrency": "string", M ISO 4217:2015
"payoutCurrency": "string", M ISO 4217:2015
"paymentMode": "string", M (B, Bank, M, Mobile Money)
"bankName": "string", 0 only Mandatory if Payment Mode is Bank
"bankBranchName": "string", 0 only Mandatory if Payment Mode is Bank
"bankBranchCode": "string", 0 only Mandatory if Payment Mode is Bank
"bankAccountNumber": "string", 0 only Mandatory if Payment Mode is Bank
"swiftCode": "string" 0 only Mandatory if Payment Mode is Bank
},
"signature": "string"
}

```

| Parameter       | Description                                                                                                | Required? |
|-----------------|------------------------------------------------------------------------------------------------------------|-----------|
| ApiKey          | API key provided for merchant account                                                                      | Y         |
| Service         | Indicates the transaction type the client wishes to initiate. <b>(List of available services provided)</b> | Y         |
| ServiceType     | Indicates whether the transaction is a credit or debit to the customer. <b>(CREDIT or DEBIT)</b>           | Y         |
| Channel         | Indicates the channel for processing the transaction. <b>(List of available channels provided)</b>         | Y         |
| CustomerRequest | Customer details for transaction                                                                           | Y         |
| InvoiceNo       | Transaction id from merchant                                                                               | Y         |

|           |                                                                                                                                                         |   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Signature | HMAC SHA256<br><br>(ApiKey+CustomerRequest.transferAmount+CustomerRequest.senderMobile+Customer+InvoiceNo)<br><br>using provided secret key for signing | Y |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|

## RESPONSE

A transaction can either be **FAILED**, **PENDING** or **SUCCESSFUL**. The status code will indicate the state of the transaction.

| Parameter     | Description                                             |
|---------------|---------------------------------------------------------|
| StatusCode    | "200":SUCCESSFUL, "300": PENDING, "404":FAILED          |
| Message       | Descriptive message on the status of the transaction    |
| TransactionId | Transaction Id generated for the request by the gateway |
| InvoiceNo     | Merchant provided transaction id                        |

The user application initiates a request to the aggregator api by accessing:

<http://<application-url>/api/payment?>

## GETBANK

Get Bank List using the api below

<http://<application-url>/api/Banks?code={{CountryCode}}>

CountryCode => 2 letter for Africa and 3 letter for Asia and Europe

```
"entryId": 2021,
"bankCode": "THAANZ",
"bankName": "ANZ BANK (THAI) PUBLIC COMPANY LIMITED",
"bankLimit": "100000",
"deliveryTime": "INSTANT",
"currencyCode": "THB"
```

## Response Parameter

| Parameter    | Description                                                      |
|--------------|------------------------------------------------------------------|
| EntryId      | <b>This is the unique ID that is passed under BankBranchName</b> |
| bankCode     | Descriptive message on the status of the transaction             |
| bankName     | Name of the Bank                                                 |
| bankLimit    | Limit to send to bank account                                    |
| deliveryTime | Time to deliver either INSTANT or T+1 or T+2                     |
| currencyCode | Currency code of Country                                         |

## GETEXCHANGERATE

You can call this api to get the exchange rate for the day.

<https://<application-url>/EZGlobalAPI/api/Rate>

```
{
  "apiKey": "string",
  "fromCountry": "string",
  "toCountry": "string",
  "currency": "string"
}
```

**CountryCode => 2 letter for Africa and 3 letter for Asia and Europe**

## Response Parameter

| Parameter | Description                                          |
|-----------|------------------------------------------------------|
| status    | <b>200 for success</b>                               |
| message   | Descriptive message on the status of the transaction |
| client    | Name of the Client                                   |
| rate      | Exchange Rate                                        |

|           |                           |
|-----------|---------------------------|
| currency  | Currency                  |
| timestamp | Time of accessing the API |

### **CALLBACKS**

All callbacks with the final status of transactions will be posted to the provided callback url with the following details:

| Parameter     | Description                                             |
|---------------|---------------------------------------------------------|
| StatusCode    | "200":SUCCESSFUL, "300": PENDING, "404":FAILED          |
| Message       | Descriptive message on the status of the transaction    |
| TransactionId | Transaction Id generated for the request by the gateway |
| InvoiceNo     | Merchant provided transaction id                        |

### **2. STATUS**

The status of every transaction can be checked by using the following url:

<http://<application-url>/api/status?apikey=&transactionid=>

| Parameter     | Description                                             |
|---------------|---------------------------------------------------------|
| ApiKey        | Api Identifier provided at the time of integration      |
| TransactionId | Transaction Id generated for the request by the gateway |

The response returned is the same as stated in the RESPONSE section above.

### **3. BALANCE ENQUIRY**

To check the balance on your account you can access the api using:

<http://<application-url>/api/balance?apikey=&service=&servicetype=>

| Parameter   | Description                                        |
|-------------|----------------------------------------------------|
| ApiKey      | Api Identifier provided at the time of integration |
| Service     | Mobile money service (eg MobileMoney)              |
| ServiceType | Wallet type ie disbursement/collection             |

#### 4. AVAILABLE SERVICES

| Service     | Country           | Available Channels | ServiceType    |
|-------------|-------------------|--------------------|----------------|
| MOBILEMONEY | Ivory Coast (CI)  | Orange             | Debit & Credit |
|             | Ivory Coast (CI)  | MTN                | Debit & Credit |
|             | Ivory Coast (CI)  | Moov               | Debit & Credit |
|             | Ivory Coast (CI)  | Wizwall            | Debit & Credit |
|             | Senegal (SN)      | Orange             | Debit & Credit |
|             | Senegal (SN)      | Wizall             | Debit & Credit |
|             | Senegal (SN)      | EMONEY             | Debit & Credit |
|             | Senegal (SN)      | Tigo               | Credit         |
|             | Burkina Faso (BF) | Orange             | Debit & Credit |
|             | Burkina Faso (BF) | YUP                | Debit & Credit |
|             | Burkina Faso (BF) | MobiCash           | Credit         |

|               |                    |               |                |
|---------------|--------------------|---------------|----------------|
|               | Mali (ML)          | Orange        | Credit         |
|               | Mali (ML)          | MobiCash      | Credit         |
|               | Niger (NI)         | AirtelMoney   | Credit         |
|               | Togo (TG)          | Moov          | Credit         |
|               | Benin (BJ)         | MTN           | Credit         |
|               | Benin (BJ)         | Moov          | Credit         |
|               | Guinea-Bissau (GB) | MTN           | Credit         |
| MOBILEMONEYGH | Ghana (GH)         | MTN           | Debit & Credit |
|               | Ghana (GH)         | Vodafone Cash | Debit & Credit |
|               | Ghana (GH)         | Airteltigo    | Debit & Credit |
| MOBILEMONEYSL | Sierra Leone (SL)  | Africell      | Credit         |
|               | Sierra Leone (SL)  | Orange        | Credit         |
| Bank          |                    | BankDirect    |                |

[illegible]



[illegible]